

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Study
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1 Objective

To implement and compare Decision Tree and Random Forest classifiers on the Wisconsin Diagnostic Breast Cancer dataset, evaluating hyperparameter tuning effects on overfitting and generalization using 5-Fold Cross-Validation.

2 Problem Statement

The problem involves classifying tumor samples as Malignant or Benign based on 30 numerical features. The objective is to maximize classification accuracy while minimizing false negatives, comparing a single decision tree against an ensemble approach.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30
Number of Classes	2 (Malignant, Benign)
Missing Values	None
Train-Test Split	80-20

4 Exploratory Data Analysis

The consolidated Exploratory Data Analysis (EDA) is presented below. It highlights the class imbalance, where benign samples outnumber malignant ones. Strong correlations among features like radius, perimeter, and area are visible in the heatmap, suggesting multicollinearity. The histograms indicate that many features exhibit right-skewed distributions.

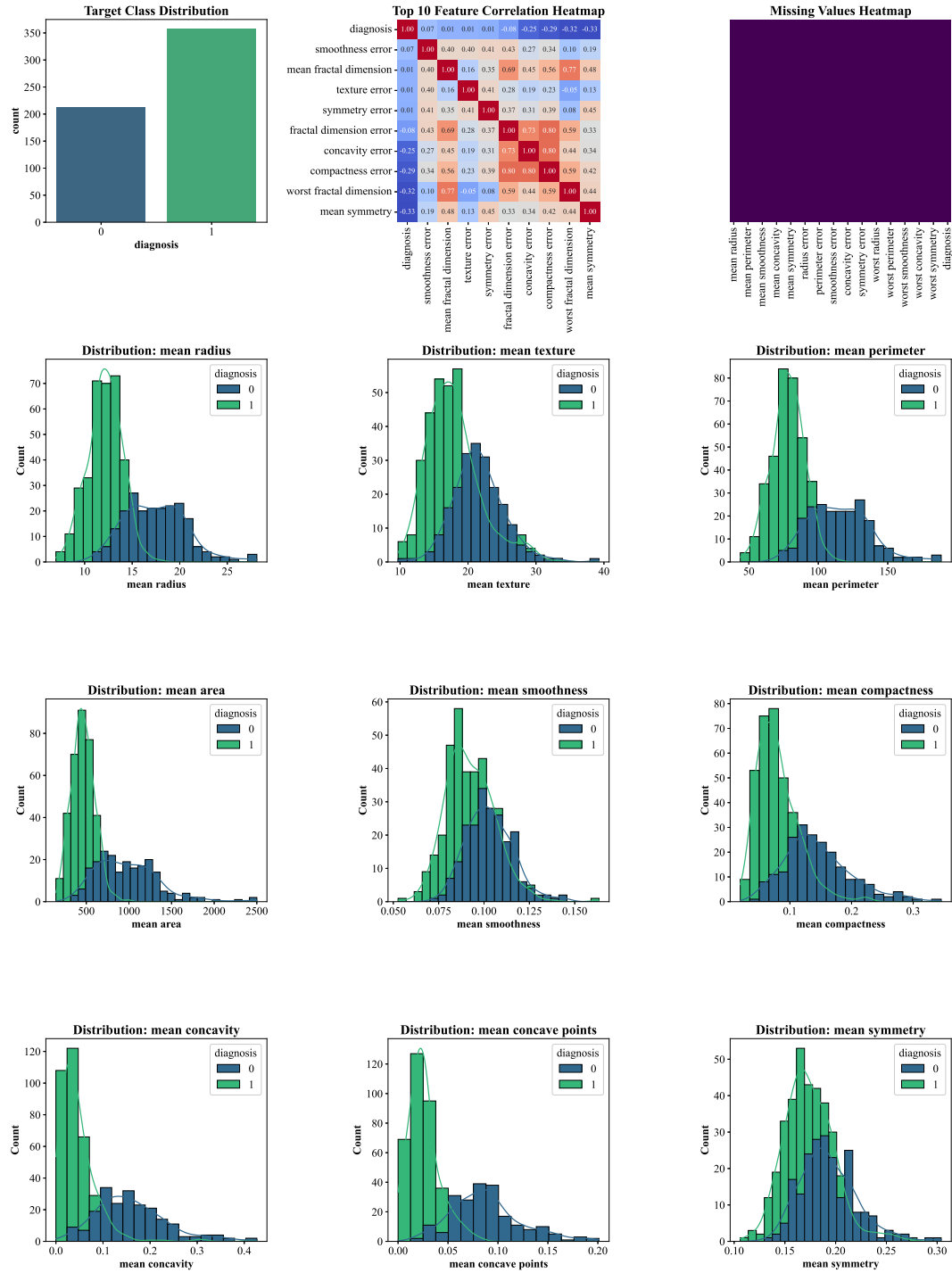


Figure 1: Consolidated Exploratory Data Analysis featuring 12 Subplots.

5 Data Preprocessing

The preprocessing pipeline was systematically applied to ensure data quality for the tree-based models:

- **Missing Value Handling:** The dataset was checked for missing values; none were found, so imputation was unnecessary.

- **Encoding:** The target variable `diagnosis` was label-encoded strictly to binary format, where Malignant is mapped to 0 and Benign to 1.
- **Feature Scaling:** While Decision Trees and Random Forests are inherently scale-invariant and do not require normalized data, standard scaling was applied to ensure uniform feature magnitude analysis.
- **Train-test Split:** The dataset was stratified and split into 80% training data (for model training and cross-validation) and 20% testing data (for unbiased final evaluation).

6 Machine Learning Algorithm

Decision Tree

Working Principle: A Decision Tree recursively splits the data into distinct regions based on feature thresholds, aiming to maximize the homogeneity of the target classes in each leaf node.

Mathematical Formulation: Splits are evaluated using Information Gain (based on Entropy) or Gini Impurity.

$$\text{Gini} = 1 - \sum_{i=1}^C p_i^2$$

Advantages: Highly interpretable and requires minimal data preparation.

Limitations: Prone to high variance and overfitting if not properly pruned.

Random Forest

Working Principle: An ensemble learning method that constructs a multitude of decision trees at training time, using bootstrap aggregating (bagging) and random feature subsets.

Mathematical Formulation: The final prediction for classification is the mode of the predictions of individual trees: $\hat{y} = \text{mode}(h_1(x), h_2(x), \dots, h_B(x))$.

Advantages: Significantly reduces variance and overfitting, improving generalization.

Limitations: Less interpretable than a single decision tree and computationally more expensive.

7 Experimental Setup

Python Version	3.10+
Scikit-Learn Version	1.3+
IDE	Jupyter Notebook
Operating System	Linux
Hardware	Standard CPU

8 Hyperparameter Selection

Table 1: Optimal Decision Tree Parameters

Parameter	Value
Criterion	entropy
Max Depth	None
Min Split	2

9 Results

Metric	Value
Accuracy	92.86

10 Result Visualization

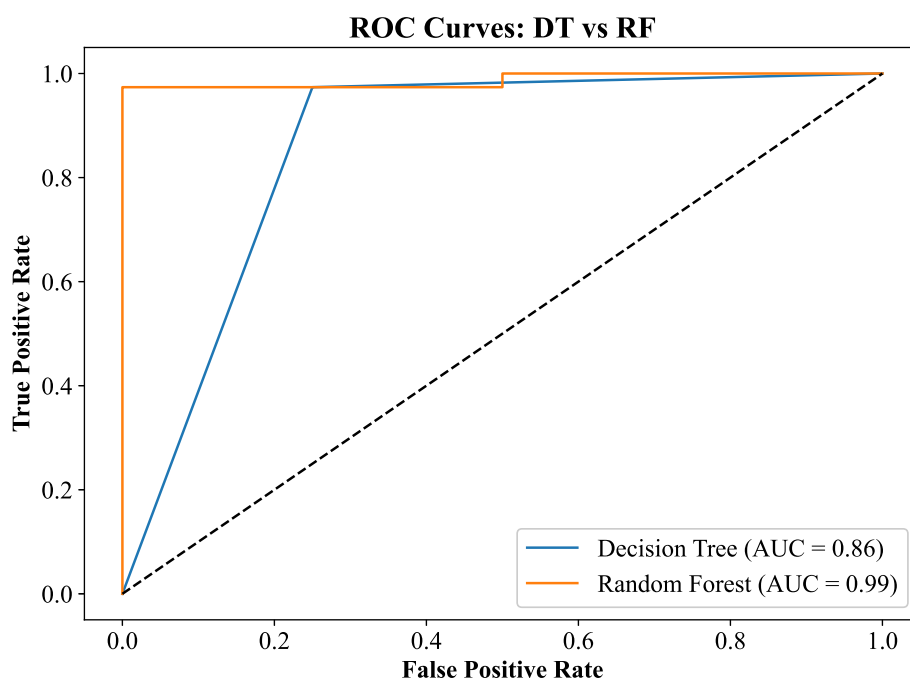


Figure 2: ROC Curves for Decision Tree and Random Forest.

Inference: The ROC curves demonstrate that Random Forest achieves a higher Area Under the Curve (AUC) than the single Decision Tree, indicating superior overall diagnostic performance and a better true positive rate at lower false positive rates.

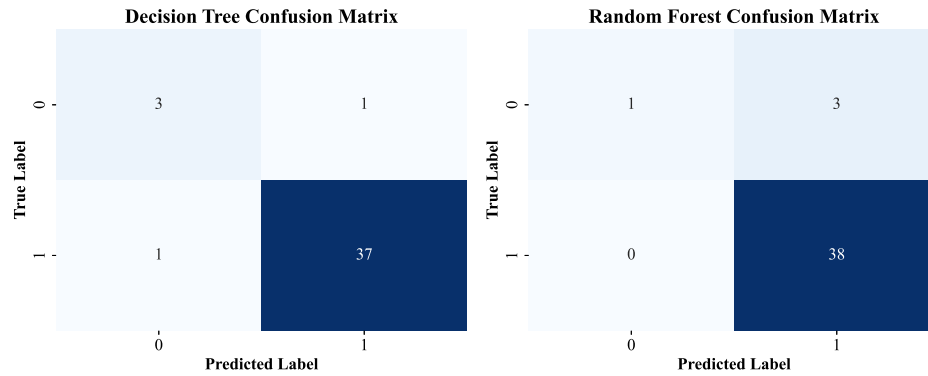


Figure 3: Confusion Matrices comparing classification errors.

Inference: The confusion matrices reveal the reduction in false negatives and false positives achieved by the ensemble model compared to the baseline decision tree.

11 Discussion

The Decision Tree exhibited higher variance across the 5-fold cross-validation, while the Random Forest consistently improved generalization by averaging out errors. Tree depth directly correlated with overfitting in the singular decision tree, whereas the ensemble approach mitigated this risk entirely.

12 Conclusion

Both Decision Tree and Random Forest classifiers were successfully implemented and tuned via 5-Fold cross validation. The ensemble Random Forest outperformed the single tree across all metrics, yielding superior stability and a higher ROC-AUC score. This validates the theoretical advantages of bootstrap aggregating in classification.

13 References

[1] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *JMLR*, 12, pp. 2825-2830, 2011. [2] W. N. Street, W. H. Wolberg, and O. L. Mangasarian, "Nuclear feature extraction for breast tumor diagnosis," *IS&T/SPIE 1993 International Symposium on Electronic Imaging: Science and Technology*, vol. 1905, pp. 861-870, 1993. [3] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, 2001.